

ADVANCED MICROECONOMETRICS
Problem Set - Day 5

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1. Consider a generalized Roy model of the form:

$$\begin{aligned} I &= Z\gamma - V \\ Y_1 &= X\beta_1 + \epsilon_1 \\ Y_0 &= X\beta_0 + \epsilon_0 \\ D &= \mathbf{1}[I > 0] \\ Y &= DY_1 + (1 - D)Y_0 \end{aligned}$$

Assume a factor structure:

$$\begin{aligned} V &= f\alpha_V + U_V \\ \epsilon_1 &= f\alpha_1 + U_1 \\ \epsilon_0 &= f\alpha_0 + U_0 \end{aligned}$$

and that U_1, U_0, U_V, f are mutually independent, with

$$U_1 \sim N(0, \sigma_1^2), U_0 \sim N(0, \sigma_0^2), \text{ and } U_V \sim N(0, \sigma_V^2).$$

Suppose that, in addition, we observe a system of external test equations regardless of D . The tests take the form

$$\begin{aligned} T_1 &= Q\theta_1 + f\delta_1 + U_{T_1} \\ T_2 &= Q\theta_2 + f\delta_2 + U_{T_2} \\ T_3 &= Q\theta_3 + f\delta_3 + U_{T_3} \end{aligned}$$

where $U_{T_i} \sim N(0, \sigma_{T_i}^2)$.

- (a) Present a formal proof of the identification of the model.

- (b) Write down the likelihood function for this model.
- (c) Assume first that $f \sim N(0, \sigma_f^2)$, that $\sigma_{U_V} = 1$, and that $\delta_1 = 1$. Why do we need to assume $\delta_1 = 1$? Could we replace this assumption with $\sigma_f = 1$? Could we have identified all the components of the model without the test equations? What if we had two time periods of Y ?
- (d) Program a maximum likelihood version of this model.
- (e) Generate a random sample of 5,000 observations under the following parameterization:

$$\begin{aligned}
f &\sim N(0, 1) \\
U_1 &\sim N(0, 1), U_0 \sim N(0, 1) \\
U_V &\sim N(0, 1), U_{T_i} \sim N(0, 1) \\
Z_0 &= X_0 = Q_0 = 1 \\
Z_1 &\sim N(0, 2), X_1 \sim N(0, 2), Q_1 \sim N(0, 1) \\
Y_1 &= 2X_0 + X_1 + 2f + U_1 \\
Y_0 &= X_0 + X_1 + f + U_0 \\
I &= 0.5Z_0 + Z_1 + f + U_V \\
T_1 &= Q_0 + Q_1 + f + U_{T_1} \\
T_2 &= Q_0 + 1.5Q_1 + 0.5f + U_{T_2} \\
T_3 &= Q_0 + 2Q_1 + 0.75f + U_{T_3} \\
D &= 1 [I > 0], \text{ and } Y = DY_1 + (1 - D)Y_0.
\end{aligned}$$

Present a table of summary statistics distinguishing between the treatment and control groups.

- (f) Estimate an OLS regression of Y ($Y_1D + (1 - D)Y_0$) on D and X . Are your results biased? Why?
- (g) Using Z as an instrument, estimate the “effect” of D on Y . Comment on your result.
- (h) Now run your OLS regression again, adding T_1 , T_2 , and T_3 as controls. What is the estimated effect of D on Y ?
- (i) Run your maximum likelihood model. You should now obtain results that estimate all the parameters correctly. Can you define a single parameter measuring the effect of moving from $D = 0$ to $D = 1$?
- (j) Define the MTE, the ATE, and the TT in this framework.

- (k) Plot the MTE as a function of $f + U_V$. Then plot the MTE as a function of V . Compare and comment on your results.
- (l) Simulate 15,000 observations from your model. Present goodness-of-fit tests.
- (m) Using the simulated data, compute the ATE and the TT. Plot your estimated TT as a function of unobserved ability.